

Tide retrospective: the multivariate identifiability bound, closing the germbij arc

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

Fifth and closing tide of the germbij arc. The germbij note proves that nonnegative real analytic losses with common compact zero locus are determined, near the locus, by the asymptotic expansions of their Boltzmann integrals against all observables; the engine is one quantitative inequality. The three-tide 1D arc landed that inequality in one dimension, and the previous tide¹ supplied the d -dimensional sector bound via scaled sets. What remained was glue: the integrated pencil identity was stated over Lebesgue on $\mathbb{R}$ only, and the composite had to be reassembled on ' $\iota \rightarrow \mathbb{R}$ '.

2 The thing formalised

Three declarations, all sorry-free. In `Laplace/Pencil.lean`: the integrated pencil identity over an arbitrary s -finite measure (`pencil_identity_integrated_measure`), whose proof is the 1D one verbatim (the Fubini lemma is generic in the second measure, and the pointwise input is the scalar identity from the first tide), and the scalar comparison `exp_pencil_ge_scalar`. In `Laplace/Multi/Identifiability.lean`: the multivariate composite.² For $L_1, L_2 \geq 0$ on ' $\iota \rightarrow \mathbb{R}$ ' with $L_1 + L_2 \leq C_0 \|w\|^2$ on the ball of radius r_0 , $g = L_2 - L_1$ bounded below at the Laplace scale on a scaled set S , and $\psi \geq 0$ equal to 1 on the ball, under $4 \leq r_0^2 t$ and explicit integrability hypotheses:

$$\text{vol}(S) \cdot c^2 e^{-4C_0} \cdot t \cdot t^{-m-d/2} \leq \int g \psi (e^{-tL_1} - e^{-tL_2}) dw, \quad d = \text{card } \iota.$$

A fixed positive power of t bounds the difference of the two Boltzmann families from below: they cannot agree to all polynomial orders. With this, the quantitative chain of germbij Theorem 7.3 is formalised in arbitrary finite dimension, with the analytic input factored into finite-vanishing-order hypotheses throughout. Numerical check (tide log): $d = 2$, $L_1 = \|w\|_2^2/2$, $g = (w_1 w_2)^2$, $t = 100$: the integral is 5.0×10^{-7} against the bound 6.1×10^{-14} .

3 Proof strategy

The composite is the 1D six-step assembly re-run: pencil identity (now the measure-general form at $\mu = \text{pi}$ Lebesgue), slice domination by the s -independent minorant via the scalar comparison, interval integration of the constant bound (slice integrability again derived from the Fubini hypothesis by product slicing), restriction of the minorant to the window $u \cdot S$ at $u = (\sqrt{t})^{-1}$, the $\psi \equiv 1$

¹2026-08-09-01-55-tide-germbij-sector-multi.

²pencil.difference.lower.bound.multi.

rewrite there, and the multivariate sector bound with $K = L_1 + L_2$, $a = g$. One improvement over the 1D version: the sector bound’s integrability hypothesis is *derived* from the minorant’s integrability by restriction and congruence on the window,³ so the multivariate composite carries no extra hypotheses relative to the 1D one.

4 Roadblocks and resolutions

Two, one notational and one infrastructural. Notational: with an explicit measure, ‘ $\int w, t * \int s \text{ in } 0..1, F \partial\mu$ ’ parses the $\partial\mu$ as the *interval* integral’s measure (a type error against ‘Measure $\mathbb{R}$ ’), so nested integrals with explicit measures must be parenthesised; after that repair everything compiled on the first attempt. Infrastructural: the previous tide’s branch turned out to be based on a stale local `main` (missing the merged 1D arc), which blocked its PR and left this tide’s chained worktree without `Pencil.lean`; one rebase onto `origin/main` (an import-list conflict) unblocked the merge and this tide proceeded on top. The lesson is recorded in the tide log: refresh the canonical clone’s `main` before every worktree creation, and verify the new worktree’s base commit.

5 What was Mathlib, what was new

Mathlib: the generic Fubini swap, product slicing, restriction and congruence for integrability, and the monotonicity lemmas already used in 1D. New: nothing conceptually; this tide’s value is completeness (the theorem now lives at the generality of the note’s statement) and the hypothesis discipline carrying over intact.

6 Lessons

Parenthesise nested integrals as soon as any measure is explicit. Verify a worktree’s base commit at creation, not at merge time. And generalising a measure-space statement is close to free when the original proof was written against generic lemmas: the 1D-to-general step here was a copy with $\partial\mu$ inserted.

7 Follow-ups

- The analytic-germ instantiation: derive the factored hypotheses (the scaled-set S with its amplitude bound) from a nonzero homogeneous leading part, connecting the Lean chain to the note’s analytic setting.
- Asymptotic packaging of the contradiction as an `Asymptotics.IsLittle0` negation.
- Deriving the integrability hypotheses from continuity plus compact support, for turnkey application to concrete potentials.

³`IntegrableOn.congr_fun` with the `EqOn` witness from $\psi \equiv 1$.